Dr. Raphael Fischer | Curriculum Vitae

AI Scientist at XplaiNLP Group | Johannes Gutenberg University Mainz
Born 10.10.1994. Nationality: German
Email: webmaster@raphischer.de
To learn about my ongoing activities, visit my webpage and LinkedIn
My publications and software are listed on Google Scholar and GitHub



My research focuses on the sustainability and trustworthiness of AI, exploring the impacts for our society, economy, and environment. As part of the XplaiNLP Group I work on narrative monitoring and disinformation detection with LLMs. My PhD contributed methods for benchmarking, reporting, labeling, and meta-learning energy-efficient ML models, accompanied by practical insights from working with Wilo. I am also a YIG Prep Pro Fellow @ KIT and voluntarily lead the Young AI Leaders Dortmund, under the UN AI for Good mission.

Professional Experience

07/26 – today **Postdoctoral Researcher** @ XplaiNLP Group @ Johannes Gutenberg University Mainz
- Verifying and extracting disinformation narratives with individualized explanations
- Coordinating the VeraXtract project from a sustainability perspective

06/19 – 06/26 **AI Scientist** @ Lamarr Institute for ML and AI @ TU Dortmund University
- Researching sustainability, resource-awareness and trustworthiness of AI
- 14 peer-reviewed publications, conference and delegation visits, transfer and network activities

02/22 – 01/25 **Data Scientist (part time)** @ Wilo SE (Dortmund)
- Working in the strategic partnership with Lamarr Institute (research transfer project)
- Exploring AI use cases and prototyping ML solutions for industrial manufacturing

07/16 – 08/18 **Working Student** @ Capgemini (Düsseldorf) & Qiagen (Hilden)
- Software development for DNA sequencing, logging and testing with Qiagen as a client

Education

07/26 – today **YIG Prep Pro Remote Fellow** @ Karlsruhe Institute of Technology
- Receiving tailored mentorship, proposal strategy guidance, and training to prepare a competitive, third-party-funded junior research group proposal

06/19 – 07/25 **PhD in Computer Science** @ TU Dortmund University – Summa cum laude | 1.0 | ECTS A
- Investigating energy efficiency, multi-objective optimization, and user perceptions in ML and AI
- Thesis topic: Advancing the Sustainability of ML and AI via Labeling and Meta-Learning
- Committee: JProf. Thomas Liebig, Prof. Geoff Webb, Prof. Christian Janiesch, Prof. Jakob Rehof

01/25 – 03/25 **Research Visit** @ Monash University, Melbourne
- Exploring resource efficiency in time series classification

10/16 – 03/19 **Master's in Computer Science** @ TU Dortmund University – 1.2 | ECTS B
- Focusing on pattern recognition, machine learning, and artificial intelligence
- Thesis topic: Filling Gaps in Incomplete Satellite Images with Compressable STRFs
- Supervisors: Prof. Katharina Morik, Dr. Nico Piatkowski

09/18 – 03/19 **Research Visit** @ Monash University, Melbourne
- Exploring probabilistic gap filling and cloud removal for satellite image series

10/13 – 09/16 **Bachelor's in Computer Science** @ TU Dortmund University – 2.4 | ECTS C
- Focusing on data visualization, computer vision, and 3D modeling

Awards & Scholarships

2026 **Nominated for the Doctoral Thesis Award from the Gesellschaft für Informatik e.V.**
- Nominated by the Computer Science Faculty at TU Dortmund University, due to an outstanding PhD

2024 **Sustainability Award of the Computer Science Faculty @ TU Dortmund University**
- Awarded for our work on Sustainable LLMs: Energy Efficiency and Transparent Labeling

2014 **Scholarship from the German Academic Scholarship Foundation (Studienstiftung)**
- For my Bachelor's and Master's studies, also volunteering as Regional Representative (06/15 – 06/17)

Teaching Experience

Supervision

01/26 – Master's student assistant Y. Abdelrahim, working on efficiency benchmarking of AI (ongoing)
03/26 – MA of T. Horny on Investigating Energy Efficiency Dimensions for Local Deployment in Large Language Models
12/25 – BA of B. Demirici on Benchmarking Large Language Models for Sustainable Evaluation
10/25 – BA of F. Meree on Interactive AI Labels for Trust: A User-Centric Reporting Framework
06/24 – Bachelor's student assistant K. Poitz, working on interdisciplinary evaluations of AI labels (until 05/26)
10/24 – BA of F. Theiner on Resource Efficiency of Adversarial Training and Inference of RobustBench Models
02/24 – BA of A. Borowski on Energy Efficiency Analysis of Knowledge Graph Embedding Models
12/22 – Master's student assistant A. van der Staay, working on benchmarking AI edge devices (until 12/23)
11/23 – BA of T. Kubiak on A Comparative Analysis of Parameter-Efficient-Fine-Tuning Methods in LLMs
09/22 – MA of F. Dillkötter on Automated Generation of Evaluation Tasks for ML Models

Lectures and Talks

05/26 – Sustainable AI Development (RC-Trust Graduate School)
04/26 – Bridging the Communication Gap: Labels for Sustainable AI Development (CEIPI AI Digital Technologies and IP Master)
04/26 – Sustainable Development in Deep Learning (ELLIS Mathematics & Efficiency of Deep Learning reading group)
10/25 – Sustainable Development in the Age of AI (Computer Science and Sustainability lecture series)
10/25 – Fundamentals of Trustworthy AI (Data, Knowledge, and Context in ML and AI lecture series)
05/25 – Sustainability and AI (Introduction to Data Science II lecture series)

Other Activities

2025 – Teaching GISSV students (remote) and organizing workshops for Digital Week Dortmund and Girls' Day @ Lamarr
2024 – Training course for teachers on AI, Environment, and Sustainability (fobizz / remote)
2023 – Organizing workshops for AI4Schools and Digital Week Dortmund @ Lamarr

Volunteering & Activities

Young AI Leaders

- Global network that empowers young enthusiasts to drive positive change with AI
- Founded and actively leading the Dortmund Hub, which has grown to 15 members
- Investigating the use of generative AI for political education in context of the Dortmund municipal elections

Scientific Community Services

- Editor for Journal of Artificial Intelligence Research (JAIR) Special Track on Challenges in Pretraining 2026
- Reviewing for International Conference on Data Mining (ICDM) 2026
- Reviewing for Conference on Fairness, Accountability, and Transparency (FAccT) 2026
- Reviewing for Explainable AI for Time Series and Data Streams Workshop (TempXAI @ ECML-PKDD) 2025 & 2024
- Reviewing for Conference on AI, Ethics, and Society (AIES) 2025
- Reviewing for Conference on Knowledge Discovery and Data Mining (KDD) 2024

- Reviewing for European Conference on Machine Learning (ECML-PKDD) 2022, 2024 & 2026

Lamarr Institute and CS Faculty @ TU Dortmund University

- Invited talks at ENFIELD Energy & Green AI Workshop (11/25), Sustainable AI Conference (09/25), Greentech.Ruhr meeting (05/25), DOPPLERS (03/25), BAuA KI-Werkstatt (11/24), AISoLA Conference (10/23), and BIFOLD All Hands Meeting (10/23)
- Posters & demos at the Trustworthy AI Summit (09/25), political visits (06/24, 09/23, 08/19), and various strategic events
- Selected delegation member for visits in Canada (10/25), USA (04/24), and France (07/22)
- Research visit & talk at Emma Strubell's Lab @ Carnegie Mellon University, Pittsburgh, USA (02/22)
- Contributed posts to the ML Blog and actively participated in lab visits, scientific forums, pitch talks, and education activities
- Sustainability Commission: Conducting and promoting sustainable efforts (06/2025 – 06/2026)
- International Coordination: Assisting students and employees with stays abroad (12/2020 – 12/2025)
- Faculty Council: Representative of the research associates (2022 – 2024)